Inflammatory Barriers

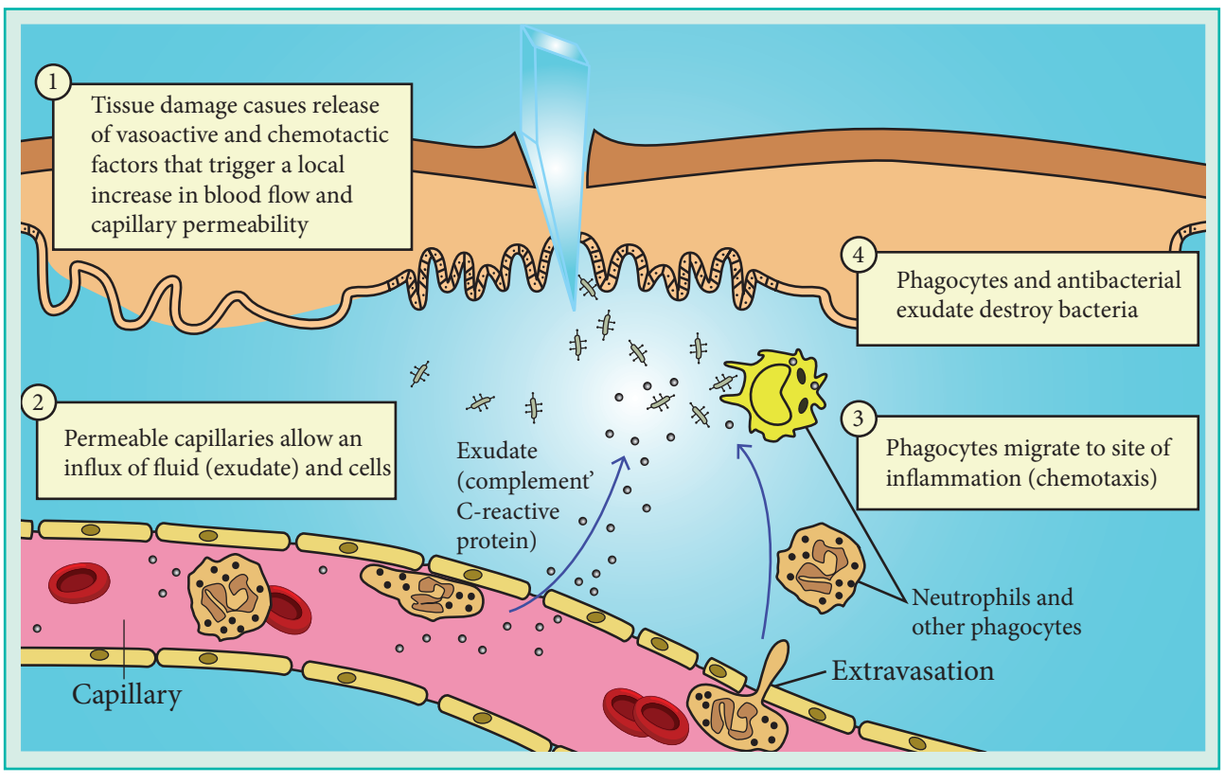
This barrier is created by the inflammatory response. Tissue damage caused by a wound or by an invading pathogenic microorganism induces a complex sequence of events collectively known as the inflammatory response. Inflammatory response is described by the "four cardinal signs of inflammation" as rubor (redness), tumor (swelling), calor (heat) and dolor (pain). Presently a fifth sign, functio laesa (loss of function),is included.

The following are the sequential steps that occur during inflammation:

Vasodilation occurs in nearby capillaries resulting in enlargement of the capillary network. The enlarged capillaries are responsible for tissue redness (erythema) and an increase in tissue temperature.

An increase in capillary permeability facilitates an influx of fluid and cells from the enlarged capillaries into the tissue. The fluid that accumulates (exudate) has much higher protein content than fluid normally released from the vasculature. Accumulation of exudate contributes to tissue swelling (edema).

Influx of phagocytes from the capillaries into the tissues is facilitated by the increased permeability of the capillaries. The emigration of phagocytes is a multistep process that includes adherence of the cells to the endothelial wall of the blood vessels (margination), followed by their emigration between the capillary endothelial cells into the tissue (diapedesis or extravasation), and finally, their migration through the tissue to the site of the invasion (chemotaxis). As phagocytic cells accumulate at the site and begin to phagocytose bacteria, they release lytic enzymes, which can damage nearby healthy cells. The accumulation of dead cells, digested material, and fluid forms a substance called pus.



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